

Atwood's Machine: Tension, Acceleration and Friction

Atwood's Machine / Atwood's Incline · oPhysics

Senior Secondary (Years 11-12)

~30 minutes

Science · Physics

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Simulation: <https://ophysics.com/f3.html>

Curriculum links: forces · motion

Learning intentions

- I can predict the direction a two-mass system will accelerate from the mass difference.
- I can apply Newton's second law to a connected two-body system to find acceleration and tension.
- I can explain how friction and incline angle change the acceleration of the system.

Getting started

Open Atwood's Machine / Atwood's Incline. Start with the incline angle set to 90° (a standard Atwood's machine) and friction switched off. Use the mass sliders to set m_1 and m_2 , then use Run, Pause and Reset to control the animation.

Tasks

1. With the incline at 90° and no friction, set $m_1 = 2$ kg and $m_2 = 1$ kg. Predict which mass will accelerate downward and calculate the expected acceleration using $a = (m_1 - m_2)g / (m_1 + m_2)$. Run the simulation and compare your answer to the displayed value.

Working:

2. Set $m_1 = m_2$ (any equal value). Predict what will happen to the system, then run it to check.
3. Keeping m_1 fixed, change m_2 to four different values and record the acceleration and string tension for each.

Value of m_2

Acceleration

Tension

4. Using your table from Task 3, describe how the acceleration changes as the mass difference ($m_1 - m_2$) increases.
5. Change the incline angle to 30° and turn friction on (a moderate coefficient). Use the free body diagram for the mass on the incline to calculate the expected tension, then compare it to the simulation's value.

Working:

Extension (optional)

Explain why the tension in the string is always less than $m_1 \times g$ when m_1 is accelerating downward, even though a stationary mass m_1 would need exactly $m_1 \times g$ of tension to hold it still.

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